

Project 1 : Hooter Module

Introduction:

Pluto X2 has a valuable number of add-ons to achieve tasks that would be otherwise not be possible. One of these add-ons is a hooter module. In this project we are going to interface it with Pluto X2.

Challenge:

Build a program to add audio generating ability to Pluto drone. Use a buzzer module to make beeping sound.

Add-On used:

Buzzer module consists of an active buzzer. It is used to generate beeps and alarms.



Connection:

Connect the Buzzer_module on the RC port on Primus V5/X2 board

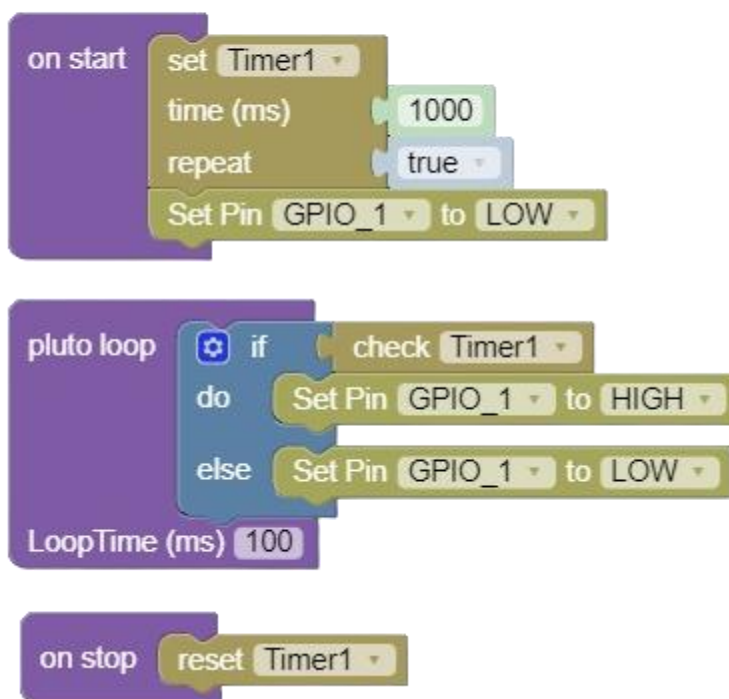
Primus V5/X2 pin	Peripheral
GND on RC port	GND on Hooter_module
Signal on RC port	VCC on Hooter_module

Approaching the Problem:

We utilise a timer, to create delay of certain milliseconds in between toggling the buzzer. This will create a beeping sound.

Solution:

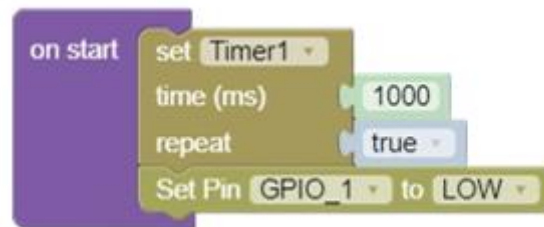
Select Receiver Mode **Rx_ESP**



Explanation of Solution:

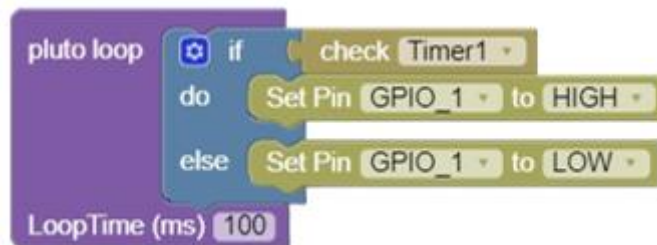
On Start Loop:

We set GPIO_1 (pin 1) to be LOW, initially and set timer of 1 seconds to be repeat



Pluto Loop:

Timer1 which is defined in On start loop, turns True, after the specified amount of time is completed. In Pluto loop we check if Timer1 is true, and upon satisfying the criteria, we make GPIO_1 HIGH, meaning turning on the buzzer. Other times the buzzer is off, GPIO_1 is LOW. This keeps repeating resulting in a beeping sound.



On stop Loop: Timer1 resets.





Fig : Pluto with Hooter Add-on

Try further!

- Utilise multiple hooter modules and create a melody.
- Add a sensor to change the Timer1's duration.
- Make text to Morse-Code generator